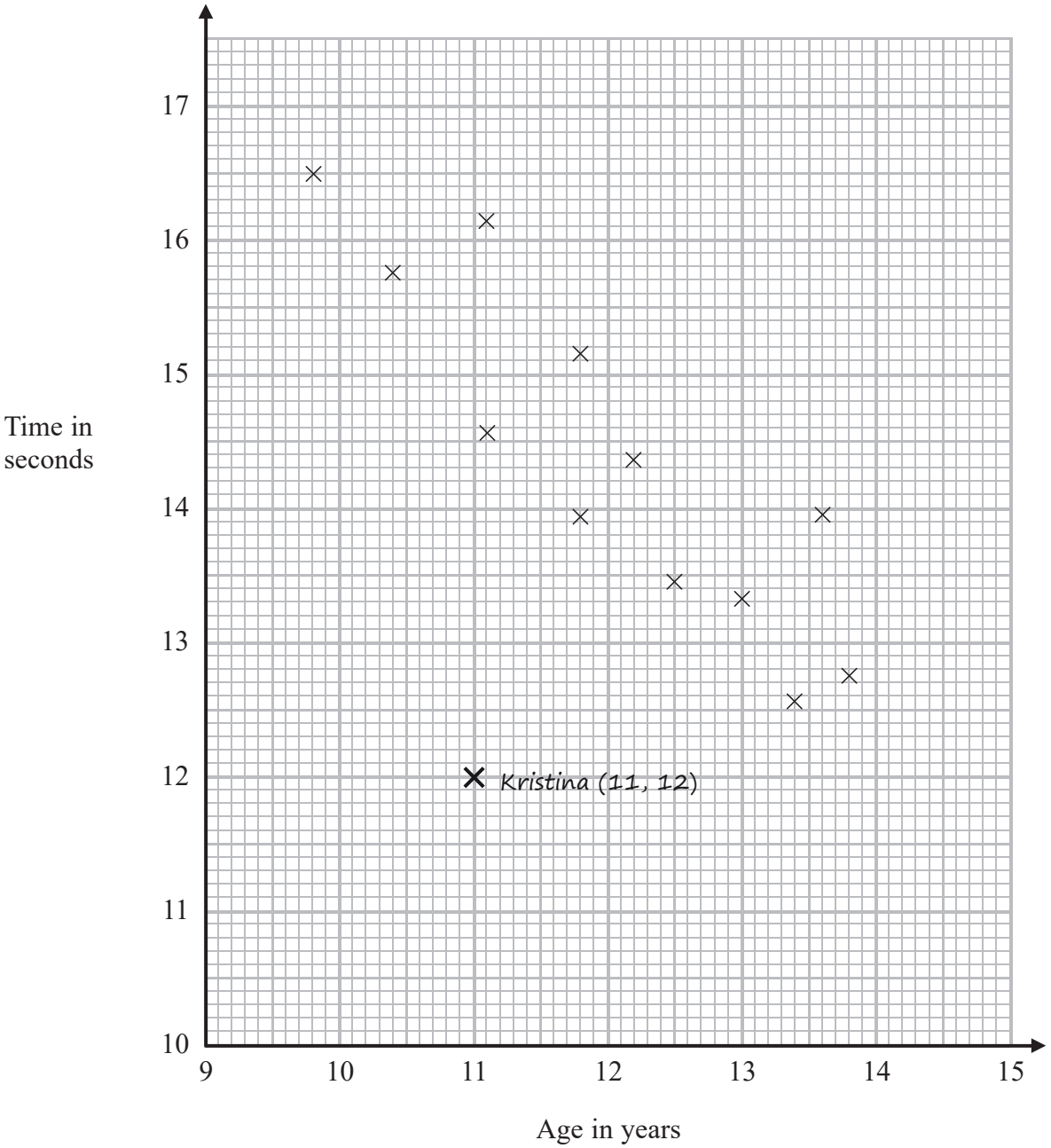
19 The scatter diagram shows information about 12 girls.

It shows the age of each girl and the best time she takes to run 100 metres.



(a) Write down the type of correlation.

Negative correlation.

(1)

1 mark
Times tend to decrease as age increases

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Kristina is 11 years old.
Her best time to run 100 metres is 12 seconds.

The point representing this information would be an outlier on the scatter diagram.

(b) Explain why.

At age 11, 12 seconds is far below the trend of the other points.

1 mark
Explain its distance from the pattern

(1)

Debbie is 15 years old.
Debbie says,

"The scatter diagram shows I should take less than 12 seconds to run 100 metres."

(c) Comment on what Debbie says.

Age 15 is outside the ages in the data, so this is extrapolation.

The graph cannot reliably support her prediction.

(1)

1 mark
Distinguish the data range from the drawn axes

(Total for Question 19 is 3 marks)

20 Expand and simplify $5(p + 3) - 2(1 - 2p)$

(Total for Question 20 is 2 marks)

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